Yinjie Zhao

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RESEARCH INTERESTS

My research interest focuses on building efficient and reliable systems for machine learning, including training, inference, and agentic workloads.

EDUCATION

University of Illinois Urbana-Champaign, Champaign, IL, United States — 2026 – Present
Ph.D. in Computer Science
Advisor: *Prof.* Francis Y. Yan

University of Michigan, Ann Arbor, MI, United States — 2024 – 2026
B.S.E. in Data Science, *Summa Cum Laude*

Shanghai Jiao Tong University - UM–SJTU Joint Institute, Shanghai, China — 2022 – 2026
B.S.E. in Mechanical Engineering

PUBLICATIONS

OpGuard: Bitwise Alignment for Precise and General Debugging of Production LLM Training
Ziming Zhou, **Yinjie Zhao**, Hang Zhu, Wenxiao Wang, Zhihao Bai, Yun Zhang, Shuguang Wang, Haibin Lin, Peng Huang
OSDI 2026 **Best Paper Nominee**

EXPERIENCE

University of Michigan – Order Lab — Ann Arbor, MI, United States
Undergraduate Research Intern – Supervisor: *Prof.* Ryan Huang — May 2025 – May 2026

OpGuard: LLM Training Debugging System (OSDI 2026)
- Developed a tensor-parallel (TP) simulator to emulate single-GPU behavior on multi-GPU setups, enabling bitwise comparison across executions with different TP configurations.
- Used OpGuard to diagnose 10 GitHub issues in projects including DeepSpeed, Megatron-LM, and Transformers.

TrainCheck: Training with Confidence (OSDI 2025) – *post-publication contributions*
- Designed and implemented the runtime checker with a watchdog to monitor the trace files.
- Designed a new proxy compatible with `torch.compile` and distributed training and reduced runtime overhead to 1.2x.
- Revised the training invariant relations for both invariant inference and checking.

Shanghai Jiao Tong University – Digital Media and Computer Vision Lab — Shanghai, China
Undergraduate Research Intern – Supervisor: *Prof.* Ran Yi — Feb. 2024 – Mar. 2025

Video Customization with Subject and Motion Control
- Designed a video customization method combining alternating fine-tuning of DreamBooth and AnimateDiff with dynamic spatial and temporal LoRA scaling.

HONORS & AWARDS

- Outstanding Graduate of Shanghai Jiao Tong University — June 2026
- Yuliming Scholarship — December 2023
- Outstanding Undergraduate Scholarship (B Level), SJTU — December 2023

TEACHING

- Teaching Assistant - CS 598 FYY (Machine Learning for Systems), UIUC — Fall 2026
- Teaching Assistant - ME2400J (Intro to Dynamics and Vibrations), SJTU — Summer 2024

LEADERSHIP & SERVICE

- Minister of the Outer Liaison Department, UM–SJTU Joint Institute Student Union — Sep. 2022 – Jul. 2024